

LIANGHONG MO

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EDUCATION

Ph.D. in Physics Department of Physics, Princeton University • M.A. in Physics, Jan. 2026. • Coursework: Reinforcement Learning (COS 435, A); Quantum Computation and Quantum Information (PHY 506, A).	Sept. 2024 – present Princeton, NJ, USA
B.S. in Physics School of Physics, Sun Yat-sen University (SYSU) • GPA: 4.4/5.0 (4.0/4.0) Ranking: 1st/79 .	Sept. 2019 – June 2024 Guangzhou, China
Summer Internship Max Planck Institute for the Physics of Complex Systems (MPIPKS)	July 2023 – Oct. 2023 Dresden, Germany

RESEARCH EXPERIENCE

Non-Markovian Corrections to Pretrained Weather Models <i>Advisor: Prof. Dmitry Abanin</i> • Many leading neural weather forecasters predict from a short window of recent states, a <i>Markovian approximation</i> in the input representation; coarse-graining can induce history dependence, motivating a correction with memory. • Augmented a frozen pretrained GraphCast with an additive correction branch — graph message passing plus Mamba layers whose latent state persists across autoregressive steps — training only the correction branch. • At 1° resolution, cut normalized weighted mean-squared error by 16.3% over ten-day forecasts and 23.7% at day ten; state-reset ablations suggest temporal memory contributes beyond added correction capacity.	Apr. – Aug. 2026
Diversity-Aware Step Weighting for RL Fine-Tuning of Diffusion Models github.com/LianghongMo/ddpo-asw • Framed reward–diversity collapse in RL-finetuned diffusion as the fixed point of the reward objective — KL regularization tilts it but does not remove it — making the <i>Pareto frontier</i> , not reward, the target. • Proposed DASW, which replaces DDPO’s uniform per-step weight with a budget-conserving weighting toward steps where generations still branch, measured by feature-space spread of predicted clean images and requiring no explicit diversity reward or occupancy penalty. • Against DDPO and simpler weighting baselines, DASW-Inception raised both LAION aesthetic reward and ImageReward on the aesthetic task while holding diversity, with qualitatively less reward hacking; DASW variants also occupied stronger Pareto-efficient regions on prompt–image alignment.	Apr. 2026 – present
Scaling Contrastive Reinforcement Learning <i>With Prof. Benjamin Eysenbach and Caleb Cheuk, Princeton University</i> • Ongoing work on scaling up contrastive RL.	Aug. 2026 – present
Cooling Algorithms for Ground-State Preparation <i>Advisors: Prof. Biao Lian and Prof. Dmitry Abanin</i> • Cast cooling-protocol design as an <i>optimal control</i> problem trading fidelity against mixing time; shaped the system–ancilla coupling via van Hove singularities to reach the Pareto frontier, beating prior protocols on gapless free-fermion, topological, and interacting systems. • Extended the protocol to adaptive particle-number control, a partially observed sequential decision problem where each cycle’s observation sets the next chemical-potential correction.	Mar. 2025 – present <i>To appear</i>

PUBLICATIONS AND PREPRINTS

- **Lianghong Mo**, Bastien Lapierre, and Qiang Miao, “Observing conformal Floquet dynamics on a digital quantum processor,” [arXiv:2605.27530](https://arxiv.org/abs/2605.27530), 2026.
- Xin-Chi Zhou*, **Lianghong Mo***, Hongzheng Zhao, and Bastien Lapierre, “Emergent Self-Similar Quantum Revivals in Spiral Drives,” [arXiv:2606.05288](https://arxiv.org/abs/2606.05288), 2026. (*equal contribution)
- Chen Bai, Xinyu Sun, **Lianghong Mo**, and Hong-Hao Tu, “Extracting central charge from ground-state overlaps of spatially deformed Hamiltonians,” [arXiv:2606.00214](https://arxiv.org/abs/2606.00214), 2026.
- Bastien Lapierre, **Lianghong Mo**, and Shinsei Ryu, “Entanglement transitions in structured and random nonunitary Gaussian circuits,” [arXiv:2507.03768](https://arxiv.org/abs/2507.03768), 2025.
- **Lianghong Mo**, Roderich Moessner, and Hongzheng Zhao, “Complex and tunable heating in conformal field theories with structured drives via classical ergodicity breaking,” [arXiv:2505.21712](https://arxiv.org/abs/2505.21712), 2025.

- **Lianghong Mo**, Zhenyu Xiao, Roderich Moessner, and Hongzheng Zhao, “Non-Hermitian delocalization in 1D via emergent compactness,” [arXiv:2412.12490](#), 2024.
- **Lianghong Mo**, Yao Zhou, Jia-Rui Sun, and Peng Ye, “Hyperfine Structure of Quantum Entanglement,” [arXiv:2311.01997](#), 2023.

TALKS

- Complex and tunable heating in driven conformal field theories. APS Global Physics Summit, Anaheim, CA, Mar. 2025.
- Cooling algorithms for ground-state preparation. APS Global Physics Summit, Denver, CO, Mar. 2026.

TEACHING AND SERVICE

- Teaching Assistant, Princeton University: PHY 103 (General Physics I / Physics Laboratory), MAT 201 (Multivariable Calculus), MAT 330 (Complex Analysis).
- Organizer, Theoretical Reading Meeting for undergraduates, School of Physics, SYSU (2020–2023); Editor, *Sap Tau Taam* journal, School of Physics, SYSU (2020–2023).

TECHNICAL SKILLS

- **Machine learning:** state-space and sequence models (Mamba), graph neural networks, diffusion and flow-matching models, policy-gradient RL (DDPO), contrastive representation learning, LoRA fine-tuning.
- **Programming:** Python (PyTorch, JAX), C, Mathematica, MATLAB, LaTeX; SLURM clusters.
- **Quantum information and many-body physics:** open quantum systems (Lindblad/Kraus), conformal field theory, quantum error correction, entanglement structure.
- **Scientific computing:** tensor networks (TEBD), exact diagonalization, Krylov/Lanczos methods, QuSpin, Gaussian fermion simulations.

HONORS AND AWARDS

- Joseph Henry Merit Award, Princeton University (2024).
- National Scholarship (Ministry of Education, China); First Prize Scholarship, SYSU (2020–2023).
- First Prize, Mathematics Competition of Chinese College Students.

LANGUAGES

- Mandarin and Cantonese (native); English (fluent); Spanish and German (basic).